

EXPERIMENT 6

PROBLEM STATEMENT

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1. Problem Statement

The primary objective of this project is to design and implement an intelligent street lighting system that automates control of street lights based on ambient light conditions. Traditional street lighting systems rely on manual switching or simple timer-based controls, which are inefficient, wasteful, and lack adaptability to real-world lighting conditions.

This project addresses these inefficiencies by deploying an STM32 microcontroller paired with a BH1750 digital ambient light sensor to create a fully automated, interrupt-driven control system. The system classifies ambient light into three operational modes — Day, Dusk, and Night — and controls street lights (via LED or relay module) accordingly, in real time.

2. Project Objectives

- **Design an interrupt-driven embedded system for intelligent street light control.**
- Interface the BH1750 light sensor with STM32 via the I2C protocol (SDA/SCL lines).
- Implement timer-based periodic interrupts (ISR) to read ambient light intensity at regular intervals.
- Classify light readings into three conditions: Day, Dusk, and Night.
- Automatically control street light output (LED/relay) based on the classification result.
- Display real-time system status via UART terminal or LCD module.
- Achieve an energy-efficient design with reduced CPU load using interrupt-driven architecture.

3. Hardware Setup

The following hardware components are integrated to form the complete system:

Component	Description	Interface
STM32 Microcontroller	Main processing unit for sensor reading and control logic	GPIO / I2C / UART
BH1750 Light Sensor	Digital ambient light intensity sensor (0–65535 lux)	I2C (SDA, SCL)
LED / Relay Module	Street light output actuator for Day/Dusk/Night control	GPIO Output
UART / LCD Display	Real-time status and classification display interface	UART / Parallel LCD
Power Supply	3.3V / 5V regulated supply for microcontroller and peripherals	VCC / GND
Connecting Wires & Breadboard	For prototyping and circuit connections	—

3.1 Circuit Connections

The BH1750 sensor is connected to the STM32 I2C bus as follows:

BH1750 Pin	STM32 Connection
VCC	3.3V Power Supply
GND	Ground (GND)
SDA	I2C SDA Pin (e.g., PB7)
SCL	I2C SCL Pin (e.g., PB6)

ADDR	GND (I2C Address: 0x23)
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4. System Operation

The system operates using a timer-based interrupt mechanism. A hardware timer is configured to trigger a periodic Interrupt Service Routine (ISR). Upon each trigger, the ISR reads the current ambient light intensity value from the BH1750 sensor over the I2C bus and classifies the reading into one of three lighting conditions.

4.1 Light Classification Logic

Condition	Lux Range	Street Light Action
Night	0 – 100 lux	Full Brightness ON (LED / Relay Active)
Dusk / Dawn	100 – 500 lux	Dimmed / Partial Light (PWM or half relay)
Day	> 500 lux	Street Lights OFF (LED / Relay Inactive)

4.2 Interrupt Service Routine (ISR) Flow

The ISR executes the following sequence on each timer interrupt:

1. Timer interrupt fires at configured periodic interval.
2. ISR initiates I2C read from BH1750 sensor.
3. Raw lux value is retrieved and processed.
4. Classification logic determines Day, Dusk, or Night condition.
5. GPIO output is set to control LED/relay accordingly.
6. Status is transmitted via UART / updated on LCD display.
7. ISR returns; CPU resumes normal operation.

5. Software Design

5.1 Firmware Architecture

The firmware is structured into the following modular layers:

- **HAL Layer: STM32 Hardware Abstraction Layer for I2C, GPIO, Timer, and UART peripherals.**
- **Sensor Driver:** BH1750 I2C driver for initialization, mode configuration, and lux reading.
- **ISR Handler:** Timer interrupt handler implementing the sensor read and classification pipeline.
- **Control Logic:** Decision module that maps lux values to Day/Dusk/Night states and drives output.
- **Display Module:** UART printf / LCD driver for real-time status reporting.

5.2 Key Software Modules

Module	Functionality
BH1750_Init()	Initialize I2C, send power-on & measurement mode command to sensor
BH1750_ReadLux()	Read 2-byte raw data over I2C, convert to lux value
TIM_IRQHandler()	Timer ISR: triggers sensor read + classification + output control
ClassifyLight(lux)	Returns Day/Dusk/Night enum based on threshold comparison
SetStreetLight(state)	Sets GPIO high/low or PWM duty cycle for LED/relay module
Display_Update(state, lux)	Sends formatted status string over UART or refreshes LCD

6. Code:

```
/* USER CODE BEGIN 4 */
void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef *htim)
{
    if(htim->Instance == TIM5)
    {
        BH1750_Read();

        if(lux > 300)
        {
            HAL_GPIO_WritePin(GPIOA,
                               GPIO_PIN_5,
                               GPIO_PIN_RESET);

            sprintf(msg,
                    "DAY | Lux = %d\r\n",
                    lux);
        }
        else if(lux > 100)
        {
            HAL_GPIO_WritePin(GPIOA,
                               GPIO_PIN_5,
                               GPIO_PIN_SET);

            sprintf(msg,
                    "DUSK | Lux = %d\r\n",
                    lux);
        }
        else
        {
            HAL_GPIO_WritePin(GPIOA,
                               GPIO_PIN_5,
                               GPIO_PIN_SET);

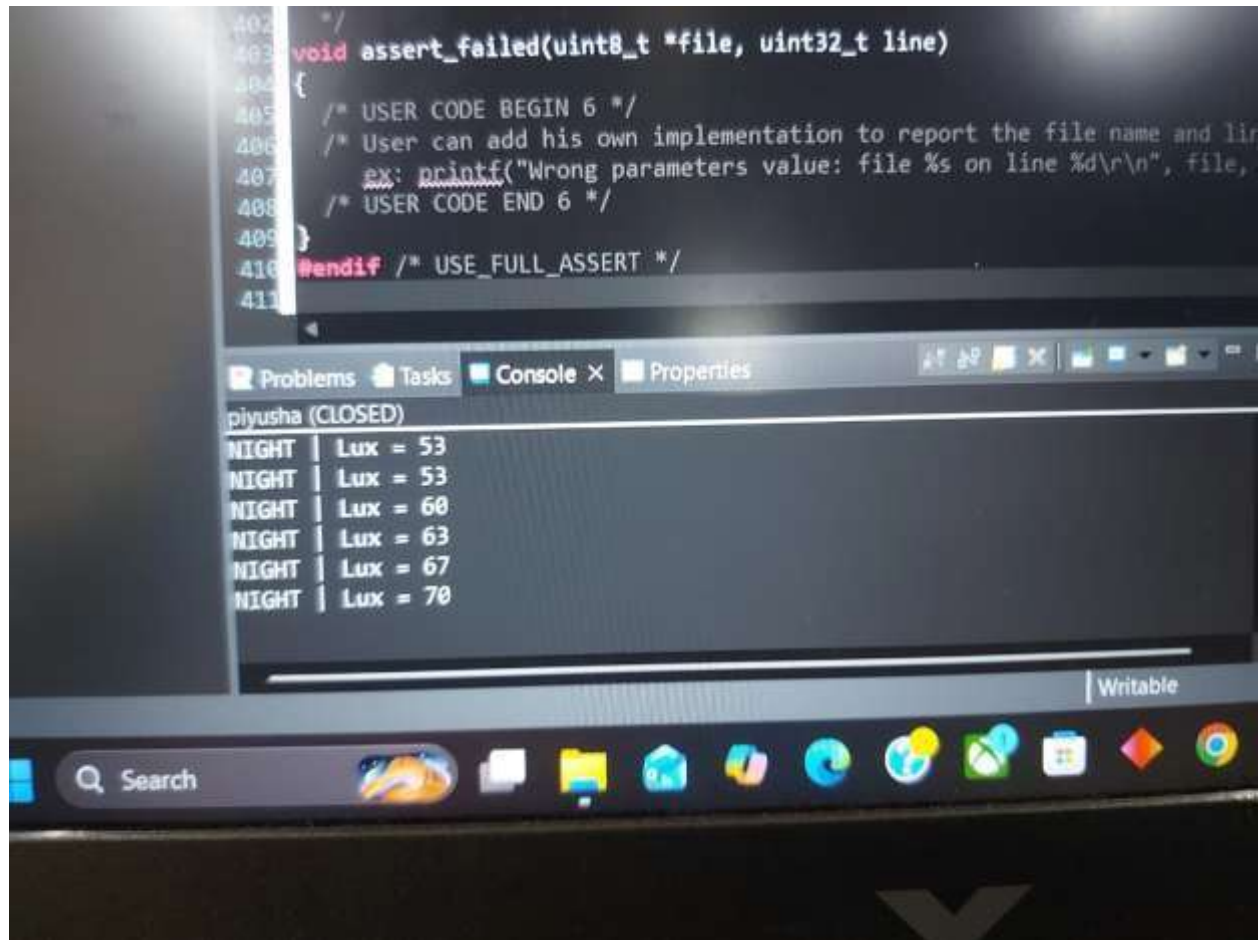
            sprintf(msg,
                    "NIGHT | Lux = %d\r\n",
                    lux);
        }

        HAL_UART_Transmit(&huart2,
                           (uint8_t*)msg,
                           strlen(msg),
                           HAL_MAX_DELAY);
    }
}
```

```
}  
/* USER CODE END 4 */
```

6. Output:





7. Advantages of the System

Advantage	Description
Energy Efficiency	Lights operate only when ambient conditions require them, reducing power consumption
Interrupt-Driven Design	Reduced CPU polling; system is highly responsive with low resource overhead
Real-Time Classification	BH1750 provides accurate digital lux values for precise Day/Dusk/Night detection

Scalability	Architecture can be extended to networks of smart street lights via wireless modules
Low Cost	BH1750 and STM32 components are affordable and widely available
Easy Integration	I2C protocol simplifies sensor connection with minimal wiring

8. Applications

- **Smart city street lighting management systems.**
- Highway and expressway automated lighting control.
- Parking lot and campus lighting automation.
- Industrial facility perimeter lighting.
- Energy-efficient building exterior lighting solutions.
- IoT-enabled smart infrastructure prototyping and research.

9. Conclusion

The PBS7 Interrupt-Driven Street Light Automation System demonstrates the effective integration of embedded systems hardware and firmware design to solve a practical urban infrastructure challenge. By leveraging the STM32 microcontroller's timer interrupt capabilities alongside the high-accuracy BH1750 digital light sensor, the system achieves fully autonomous, energy-aware street lighting control.

The interrupt-driven approach ensures that system responsiveness is maintained without burdening the CPU with continuous polling cycles, making this design both efficient and scalable. The real-time status output via UART/LCD further enhances the system's transparency and usability. This project serves as a strong foundation for more advanced smart city lighting deployments and embedded IoT applications.

End of Report — PBS7 Interrupt-Driven Street Light Automation System